Feasibility Study

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study.

The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below:

- To analyse whether the software will meet organizational requirements
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule
- To determine whether the software can be integrated with other existing software

If a system does not support the business objectives, it has no real value to the business. While this may seem obvious, many organisations develop systems which do not contribute to their objectives either because they don't have a clear statement of these objectives, because they fail to define the business requirements for the system or because other political or organisation factors influence the system procurement.

Information Assessment

The information assessment phase identifies the information that is required to answer the key feasibility questions. Once the information has been identified, information sources are questioned to discover the answers to these questions. The following questions were asked to evaluate the feasibility of the LittleLumin platform:

	Question	Response
1	How would the organization cope if this system was not implemented?	Parents would continue to rely on screen-based educational apps for their children, which contribute to excessive screen time and passive learning. Occupational therapists and child specialists would lack remote monitoring tools, limiting their ability to support families between sessions. Early intervention for developmental delays would remain delayed due to lack of accessible screening and tracking tools.
2	What are the problems with current processes and how would a new system help alleviate these problems?	Current solutions either place children directly on screens (causing health and developmental concerns), provide generic one-size-fits-all content that ignores individual learning patterns, or fail to address parental wellbeing. LittleLumin solves these problems by offering personalized, screen-free activities generated in real-time based on each child's unique learning pattern, tracking development using clinically validated tools (VABS-II), and providing emotional wellbeing support for parents.
3	What direct contribution will the system make to the business objectives and requirements?	The platform directly contributes to improving child developmental outcomes through personalized therapeutic activities. It supports caregiver wellbeing through mood tracking and emotional support. It provides professionals with remote monitoring capabilities and data-driven insights. It also contributes to reducing screen time for young children, which is a growing public health concern.
4	Can information be transferred to and from	Yes. The platform is built using Firebase Firestore which supports integration with external systems. It can connect with OpenAI API and HuggingFace API for AI-powered content

	Question	Response
	other organizational systems?	generation. In the future, it can be extended to integrate with Electronic Health Record (EHR) systems, wearable devices, and school management systems through RESTful APIs.
5	Does the system require technology that has not previously been used in the organization?	The organization currently does not have an AI-powered therapeutic platform. However, the technologies selected for this project—Flutter (Dart), Python FastAPI, Firebase, OpenAI API, and OpenCV—are all well-established, stable, and widely adopted technologies with extensive documentation and community support. The development team has experience with most of these technologies.
6	What must be supported by the system and what need not be supported?	The system must support personalized activity generation based on developmental needs, developmental milestone tracking using VABS-II, image-based handwriting analysis, caregiver wellbeing tracking, and professional dashboards for monitoring. The system will not support clinical diagnosis, replacement of licensed healthcare professionals, or medical treatment. It is a supportive, non-diagnostic tool.

Types of Feasibility

Various types of feasibility that are commonly considered include technical feasibility, operational feasibility, and economic feasibility.

3.1 Technical Feasibility

Technical feasibility assesses the current resources (such as hardware and software) and technology, which are required to accomplish user requirements in the software within the allocated time and budget. Technical feasibility also performs the following tasks:

- Analyses the technical skills and capabilities of the software development team members
- Determines whether the relevant technology is stable and established
- Ascertains that the technology chosen for software development has a large number of users so that they can be consulted when problems arise or improvements are required

Technology Readiness:

The development team possesses knowledge of Flutter (Dart) for frontend development, Python for backend development, and Firebase for database management. Any additional learning required for OpenCV and Tesseract OCR is manageable through available online resources.

Flutter is a mature cross-platform framework backed by Google. FastAPI is a modern high-performance Python framework. Firebase is a widely adopted backend platform. OpenAI API and HuggingFace API are industry standards for AI integration. All selected technologies have large active communities with extensive documentation and support.

Hardware Requirements:

The hardware requirements include a development machine with Windows, Mac, or Linux operating system, 8GB or more RAM, and 256GB or more storage. Mobile testing can be done on Android (API 21+) and iOS (12+) devices or emulators. The infrastructure is cloud-based using Firebase, requiring no physical server. A stable internet connection is required for API calls and cloud services.

Software Requirements:

The software requirements include Flutter 3.x and Dart 3.x for frontend development, Python 3.10+ and FastAPI 0.100+ for backend development, Firebase Firestore for database, Firebase Authentication for secure access, OpenAI API and HuggingFace API for AI integration, OpenCV 4.x and Tesseract OCR 5.x for image processing, ReportLab 4.x for PDF generation, Git and GitHub for version control, and Visual Studio Code or Android Studio as the development environment.

The LittleLumin platform is technically feasible because all technologies used are mature, stable, and well-documented, the development team has the required skills and can acquire additional knowledge as needed, hardware and software requirements are minimal and affordable, and the development timeline of 8 months is realistic for the proposed scope.

3.2 Operational Feasibility

Operational feasibility assesses the extent to which the required software performs a series of steps to solve business problems and user requirements. This feasibility is dependent on human resources (software development team) and involves visualizing whether the software will operate after it is developed and be operative once it is installed. Operational feasibility also performs the following tasks:

- Determines whether the problems anticipated in user requirements are of high priority
- Determines whether the solution suggested by the software development team is acceptable
- Analyzes whether users will adapt to a new software
- Determines whether the organization is satisfied by the alternative solutions proposed by the software development team

Problem Priority:

The problems addressed by LittleLumin are of high priority. Excessive screen time for toddlers is a growing concern with 80% of parents giving phones to toddlers and WHO recommending zero screen time for under 2s. Lack of parenting guidance affects 80% of parents who feel unprepared for their role. The absence of support for caregiver mental health is critical as 1 in 7 parents experience post-partum depression. One-size-fits-all educational content fails to address individual learning patterns, and therapists lack remote monitoring tools to support families between sessions**.**

User Acceptance:

Parents will adopt the platform because it provides quick, easy, and structured guidance—saving time and reducing stress. Children will benefit through play-based learning with no screen required. Teachers will gain insights into student progress and identify children needing support. Occupational therapists will gain remote monitoring capabilities and data-driven

insights. Child psychologists will track patient progress and provide recommendations remotely. The platform does not replace professionals but supports and extends their reach.

The LittleLumin platform is operationally feasible because the problems it addresses are of high priority, the solution has been validated through research and stakeholder feedback, users are likely to adopt the platform due to its simplicity and clear value, and the platform supports professionals rather than replacing them.

3.3 Economic Feasibility

Economic feasibility determines whether the required software is capable of generating financial gains for an organization. It involves the cost incurred on the software development team, estimated cost of hardware and software, cost of performing feasibility study, and so on.

- Cost incurred on software development to produce long-term gains for an organization
- Cost required to conduct full software investigation (such as requirements elicitation and requirements analysis)
- Cost of hardware, software, development team, and training

Cost Estimation:

As a student project, there are no salary costs for the software development team. Existing laptops and mobile devices will be used for development, incurring no hardware costs. All software tools used are open-source or have free tiers available. Firebase services are available for free initially. OpenAI API costs are pay-as-you-go based on usage, while HuggingFace API offers a free tier for most models. Cloud hosting costs are minimal, with free tiers available for initial deployment. Training can be done through freely available online resources. A small contingency amount is kept for any unexpected costs.

Cost-Benefit Consideration:

The platform offers significant value in terms of improving child developmental outcomes, supporting caregiver mental health, enabling professional remote monitoring, and reducing screen time for young children. The platform provides scalable support for parents and professionals, complementing existing therapeutic services.

The LittleLumin platform is economically feasible because development costs are minimal as a student project, operating costs are low, and the platform offers significant value in terms of social and professional benefits.